Data Structures & Algorithms — Weekly Assessment

Student ID: 2303031241262

Question:
Explain the working principle and derive the exact time complexity of Binary Search.

Student Answer:
Binary search works by looking at every element in an array sequentially until it finds the key. Because it checks each element from index 0 to $n-1$, the recurrence is $T(n) = T(n-1) + 1$, which solves to $O(n)$ in all cases. It can be applied directly on any unsorted list without preprocessing.